TN LandMate: PDF Architecture & Workflow Working Flow

Complete End-to-End System Design & Operation Guide (In Tanglish)

v1.0.0 Production

> **Vanakkam!** Namma **TN LandMate** app-la PDF generation, live scraper bridge, math calculations, and offline storage epdi synchronize aaguthu nu indha document-la detailed-ah step-by-step explain pannirukom.
>
> Indha system **5 Core Architectural Layers**-la build aagirukku: Offline Geography, Live Govt Bridge, Math Normalization, Vector PDF Engine, and Native Device Spooler.

1. Visual Architecture Blueprint (Architecture Flow)

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[1. User Input Form] | +---> [Local Geography DB: 38 Districts, 317 Taluks, 18,606 Villages] (100% Offline) | +---> (Optional) [Live Govt Bridge → FastAPI Backend → eservices.tn.gov.in] | +⟶ Live CAPTCHA solved in Native Mobile BottomSheet +⟶ Extracted Official Records (Owner Name, Survey, Land Class) | [2. Data Normalization & Math Engine] | +---> Converts Extent: Cents → Sq.Ft (x435.6) → Hectares → Ares +---> Generates Unique Reference Ref: TN-REV-2026-XXXXXX +---> Calculates Stamp Duty & Registration Tax Assessment | [3. Client-Side PDF Engine (PattaPdfService)] | +---> Builds A4 Vector Canvas (Header, Bilingual Tables, Statutory Footer) +---> Embeds Live Vector QR Code targeting eservices.tn.gov.in validation +---> Section 65B Indian Evidence Act / IT Act 2000 Legal Validity Stamp | [4. Native Mobile Device Actions (PattaPreviewScreen)] | +---> Save as PDF into phone's Downloads directory +---> Direct Share via WhatsApp, Gmail, Telegram +---> Direct Print to WiFi / Bluetooth Network Printers
```

2. The 5 Core Layers Deep-Dive (In Tanglish)

1 Layer 1: Zero-Latency Offline Geography Engine

100% Offline

> **Enna Nadakuthu?** User app open panni District, Taluk, Revenue Village select pannum pothu internet connection thevaiye illa! Tamil Nadu-oda 38 Districts, 317 Taluks, and 18,606 Villages mobile memory kulla embed aagi irukku.

Technical Source: `mobile/lib/core/data/tn_geography_data.dart`. Instant binary lookup with zero network lag.

2 Layer 2: Live Government Portal Scraper Bridge

Cloud + Govt Live

> **Enna Nadakuthu?** User "Extract Live Patta/Chitta" click panna, namma Render backend direct-ah `eservices.tn.gov.in` kooda connect aagi, live session create panni, CAPTCHA image-ah phone screen-la kaatum. User solve panna udane official Tamil Nadu server data retrieve aagi varum.

Technical Source: `backend/app/services/live_scraper_service.py` & `mobile/lib/shared/widgets/live_gov_extractor_sheet.dart`. 60s cold-start timeout tolerance.

3 Layer 3: Math & Normalization Engine

Client & Server

> **Enna Nadakuthu?** Raw land extent-ah 4 standard revenue units-ku automatically convert pannum: Cents, Square Feet, Hectares, and Ares. Unique timestamped reference ID generate aagi certificate seal create aagum.

Formula: `Sq.Ft = Cents * 435.6` | `Hectares = Cents * 0.00404686` | Reference: `TN-REV-YYYY-XXXXXX`.

4 Layer 4: Vector PDF Generation Engine (PattaPdfService)

100% Client-Side Offline

> **Enna Nadakuthu?** Server illamalaye mobile phone-oda RAM-la official Tamil Nadu Revenue Department A4 certificate format-la pixel-perfect PDF build aagum. Vector QR code, bilingual tables, and IT Act Section 65B legal validity stamp embed aagidum.

Technical Source: `mobile/lib/core/services/patta_pdf_service.dart` using Dart `pdf` & `printing` libraries.

5 Layer 5: Native Device Integration (Preview, Save & Share)

Native Android/iOS

> **Enna Nadakuthu?** Generated PDF-ah user phone-la zoom panni paakalaam, 1-tap panni phone-oda Downloads folder-la save pannalaam, WhatsApp-la clients-ku share pannalaam, illana Bluetooth/WiFi printer vazhiya print edukalaam!

Technical Source: `mobile/lib/features/records/patta_preview_screen.dart`. Uses Native Android Print Spooler and Intent Share Sheet.

3. End-to-End Real World Citizen Journey (Tanglish)

> **Scenario:** Oru real estate buyer illana land owner Ambattur Survey No 104/2B-ku Patta copy edukka virumburaanga:
> 1. **Open App:** Home screen-la *"Download Patta PDF"* button tap panraanga.
> 2. **Instant Select:** District: *Chennai*, Taluk: *Ambattur*, Village: *Ambattur* select panraanga (offline-la instant-ah load aagum).
> 3. **Data Input:** Survey No *104*, Subdiv *2B*, Patta No *1420*, Owner *R. Selvam* kudukuraanga.
> 4. **1-Tap Generate:** *"Download Generated Patta (PDF)"* tap panna udane microsecond-la official A4 PDF certificate generate aagi screen-la open aagidum.
> 5. **Download / Share:** Top right download/share button thatti WhatsApp-la bank loan officer-ko buyer-ko instant-ah send panraanga!

4. Architecture Summary Table

Module	Key Responsibility	Offline Capable?	Primary File
Geography DB	38 Districts, 317 Taluks, 18,606 Revenue Villages	Yes (100%)	`tn_geography_data.dart`
Live Govt Bridge	Connects to eservices.tn.gov.in & extracts live records	Requires Internet	`live_scraper_service.py`
PDF Generator Engine	Builds official A4 Patta with QR Code & IT Act seal	Yes (100%)	`patta_pdf_service.dart`
Preview & Spooler	Interactive zoom, storage download, WhatsApp share, print	Yes (100%)	`patta_preview_screen.dart`
Cloud Backend API	Cloud metadata validation, guideline values & health	Requires Internet	`land.py`